

STILLWATER 2 W, OK, USA

WMO: 746420

Lat: 36.1181N

Lon: 97.0914W

Elev: 890

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	17.2	1.1	6.0	15.5	5.8	7.6	20.7	19.0	38.8	16.9	39.5	5.7	N/A	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	100.9	73.9	97.6	74.2	94.7	74.3	78.1	92.2	77.2	91.1	76.2	89.8	6.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	132.2	85.0	73.2	127.5	84.1	72.2	123.1	83.2	42.2	92.1	41.1	91.3	40.1	89.9	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	13.9	12.2	3.7	104.0	7.0	3.6	-1.4	106.6	-5.5	108.7	-9.4	110.7	-14.5	113.3
			1.2	79.7	8.4	1.5	-4.9	80.8	-9.8	81.6	-14.5	82.5	-20.7	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	37.9	40.7	52.0	60.2	68.9	78.3	82.4	81.3	73.6	61.1	49.8	39.8
	DBStd	18.05	10.50	11.48	10.77	9.22	8.07	5.41	5.49	5.82	7.06	9.40	9.46	9.74
	HDD50	1289	395	299	103	20	2	0	0	0	0	17	116	337
	HDD65	3643	840	682	421	197	55	1	0	0	13	189	463	782
	CDD50	5161	19	38	166	327	586	849	1004	972	709	361	109	21
	CDD65	2041	0	2	18	55	175	400	539	507	272	67	6	0
	CDH74	23342	4	21	162	572	1702	4480	6765	6053	2830	686	65	2
	CDH80	11270	0	3	29	149	591	2084	3717	3250	1240	199	8	0
Wind	WSAvg	5.1	5.5	5.9	6.3	6.4	5.3	4.9	4.3	4.2	4.2	4.6	5.0	5.1
Precipitation	PrecAvg	34.6	1.2	1.4	2.7	3.1	4.9	4.3	3.0	3.1	4.1	3.0	2.3	1.6
	PrecMax	46.4	3.7	5.4	7.7	8.3	14.6	10.6	8.2	8.5	12.4	8.7	6.7	5.1
	PrecMin	19.7	0.0	0.1	0.0	0.2	0.6	0.8	0.0	0.0	0.0	0.4	0.0	0.1
	PrecStd	6.7	1.1	1.2	1.9	1.9	2.8	2.4	1.9	2.1	2.6	2.1	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.7	77.1	83.8	89.2	93.0	99.9	105.2	106.2	98.3	89.3	80.5	70.9
		MCWB	53.4	58.2	62.9	66.8	70.7	74.9	72.7	72.8	71.6	70.6	62.0	56.0
	2%	DB	67.1	71.2	78.4	84.4	89.5	95.6	101.5	100.8	93.4	85.5	75.3	65.6
		MCWB	52.2	56.4	60.3	63.9	70.5	74.5	73.3	74.6	72.2	66.7	59.5	55.0
	5%	DB	61.8	65.9	74.3	80.4	86.5	92.8	97.9	97.4	90.7	81.6	71.0	60.4
		MCWB	50.0	54.7	58.4	62.2	69.4	74.4	74.3	74.6	72.3	65.1	58.0	49.9
	10%	DB	56.1	60.3	70.0	76.8	83.1	90.3	94.9	94.1	87.3	77.5	66.5	56.6
		MCWB	45.4	50.2	56.5	62.4	68.6	73.8	74.9	73.8	70.8	63.8	55.2	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	62.4	67.2	69.9	75.8	78.8	80.1	79.0	76.7	72.9	66.8	62.3
		MCDB	64.4	70.3	79.0	80.8	87.1	93.5	92.8	94.5	89.7	84.8	74.1	68.4
	2%	WB	52.9	59.6	64.3	67.8	73.5	77.6	78.3	77.7	75.1	71.1	62.3	56.3
		MCDB	62.6	68.7	75.2	78.5	84.9	91.2	91.6	93.4	88.8	81.1	70.1	60.1
	5%	WB	49.4	56.1	61.7	66.2	72.0	76.6	77.2	76.6	74.0	69.2	59.8	52.1
		MCDB	60.2	64.7	71.5	76.9	82.8	89.6	90.6	91.5	87.1	78.0	68.0	58.9
	10%	WB	45.8	50.7	58.2	64.2	70.4	75.3	76.1	75.4	72.9	66.8	56.9	47.9
		MCDB	54.4	60.8	67.5	74.2	80.4	87.8	89.8	89.2	84.9	74.7	64.0	55.0
Mean Daily Temperature Range	5% DB	MDBR	22.8	22.3	23.4	22.9	20.7	19.9	20.9	21.4	22.3	23.1	23.5	21.2
		MCDBR	33.1	31.0	28.3	27.7	23.8	22.7	25.0	25.9	24.8	27.2	29.2	28.7
		MCWBR	19.7	18.8	15.3	14.0	10.0	7.6	6.6	7.3	8.6	12.1	16.1	18.3
	5% WB	MCDBR	30.1	27.3	23.5	22.9	19.7	19.7	20.1	21.2	20.2	20.3	22.1	23.3
		MCWBR	19.3	18.4	13.5	13.6	9.9	7.8	7.3	7.9	7.7	11.7	15.3	17.6
Clear-Sky Solar Irradiance	taub	0.297	0.302	0.342	0.382	0.419	0.429	0.434	0.436	0.393	0.348	0.314	0.291	
	taud	2.496	2.494	2.411	2.315	2.284	2.291	2.316	2.254	2.358	2.470	2.504	2.524	
	Ebn at Noon	287	299	294	286	275	271	269	266	273	278	279	283	
	Edh at Noon	26	29	35	40	42	42	41	43	36	30	25	24	
All-Sky Solar Radiation	RadAvg	850	1049	1370	1704	1870	2136	2139	1935	1609	1211	902	718	
	RadStd	84	157	87	118	189	190	118	118	134	149	113	77	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page